POST-SCREEN HUMAN EXPERIENCE INFRASTRUCTURE

Embodied Presence Internet

Upload CSV

Rule-Based Insight · experimental prototype

Session P-002 showed high workload is paired with elevated error rates.

Refresh

MISSION CONTROL / ● LIVE

P-001

Remote object assembly

VR headset + haptic glove

68.8

rule-based stability indicator

94
Stress

77
Performance

DEGRADED

100
Latency

Heuristic embodiment state

73
Agency

74
Presence

52
Ownership

4:19:16 PM / 119ms / heuristic stability 69% / rule flag: embodiment degradation

PDF-DERIVED EVALUATION FRAMEWORK

CATEM Cross-Layer Assessment

evidence-condition profile

Experience **69.3**	Action **74**
Medium · 100% Coverage	Medium · 100% Coverage
Human state & cognition **55.3**	System conditions **68.6**
Medium · 100% Coverage	Medium · 100% Coverage

Dt **Data & interpretation conditions** **81.5**

Cross-cutting record metadata · not an observational layer

high · 100% coverage

Metric coverage	Synchronization	Missing data
100 %	**83.2 %**	**3.2 %**

SELECTED EVIDENCE WINDOW

Robot object drop

-2s to +2s · synthetic fixture

Layer / measure	-2s	-1s	0s	+1s	+
EXPERIENCE **Agency**	78	72	**58**	64	7
ACTION **Task error**	0	0	**1**	0	
HUMAN STATE **Workload**	58 %	64 %	**82 %**	72 %	6
HUMAN STATE **Heart rate**	88 bpm	96 bpm	**112 bpm**	104 bpm	93
SYSTEM **Latency**	96 ms	118 ms	**220 ms**	154 ms	10
SYSTEM **Packet loss**	1.2 %	2.4 %	**12 %**	5.1 %	1.

Layer / measure	-2s	-1s	0s	+1s	+
SYSTEM **Tracking dropout**	2 %	3 %	**15 %**	8 %	3

> The five samples are deterministic authored values for software verification. Temporal co-occurrence does not establish causality.

CATEM Propositions P1–P6

live cross-layer tests

P1 MONITOR

Latency asymmetry

119 ms latency; agency 73; ownership 52

P2 MONITOR

Presence–performance dissociation

presence 74; workload 65; errors 8.9%

P3 MONITOR

Assistance trade-off

assistance 44; agency 73; trust 71

P4 MONITOR

Physiological leading indicators

HRV 50 ms; GSR 0.34; jitter 9 ms

P5 OBSERVED

Synchrony over fidelity

behavioral synchrony 72; avatar fidelity 66

P6 SUPPORTED

Ethical adaptation

disclosed True; overridable True; trust 71

Agency-Preserving Adaptation

explainable and overridable

- Reduce rendering complexity and protect agency.
- Continue monitoring workload.
- Adaptation disclosure is active.

Experimental Dashboard Modules

future-work prototype

> The modules below use deterministic rules or heuristic composites. They are not validated CATEM paper contributions, predictive models, diagnostic tools, or evidence of causal relationships.

Live Heuristic Streams

1 frames buffered

Latency **119ms**

Rule-based stability indicator **69%**

Heuristic embodiment estimate **45%**

Rule-based risk signal **61%**

Synthetic Telemetry Simulator

baseline alignment

Physiology

HR
94.43 bpm

HRV
50.09 ms

Stress
43.29 %

Network

Latency
118.94 ms

Jitter
9.2 ms

Loss
3.25 %

Embodiment

Agency
72.74 %

Ownership

Stress Escalation

Workload
64.97 %

Errors

51.63 %

8.93 %

Degrade

33.85 %

Timing

25.8 ms

Multimodal Presence Panel

partial-presence

Presence

68.3 %

Intention

73.9 %

Emotion

78.6 %

Embodiment

44.9 %

Adaptive Support Panel

experimental · rule-based · calm

Presence is holding. I will maintain sensory fidelity and monitor for drift.

- retain current rendering settings
- review network conditions
- retain current haptic settings

Longitudinal Session Profile

76.1% continuity

P-001 adapts as a needs reinforcement with haptic reinforcement as the current recovery path.

Stability mode	**performance**
Preferred recovery	**haptic reinforcement**

Full-Sensory Telepresence

65% fidelity

Haptics

48.7 %

Visual

81.4 %

Audio

Tactile

79.8 % | 50.1 %

Shared Reality Space

co-present

Trust **77.3 %**	Team load **48.8 %**
Social sync **77.3 %**	Augment **75.1 %**

Adaptation Control Panel

experimental · rule-based · stabilize

preserve embodiment continuity

- retain current rendering settings
- review network conditions
- retain current haptic settings
- rebalance sensory fidelity

Embodiment	71.7	Medium
Presence	75.5	High
Performance	78.1	High
Behavior	72.4	Medium
Physiological	53.4	Medium
System	65.5	Medium
Visualization	50.5	

Medium

Real-Time Multimodal Stream

real_telemetry_simulator

Latency **118.94 ms**	Packet loss **3.25 %**
Heart rate **94.43 bpm**	Agency **72.74**

Heuristic embodiment estimate **44.9**

degraded

Cognitive State Heuristic

experimental · low rule-based flag

Stability **68.8 %**	Rule activation index **36.2**
Attention drift **29.2 %**	Stress **nominal**

Rule-Based Risk Review

experimental · descriptive threshold logic

Nominal Input Condition

No event timing or probability is estimated.

- retain current rendering settings
- review network conditions
- retain current haptic settings

> Deterministic threshold review only; no event time, probability, predictive accuracy, diagnosis, or causal effect is estimated.

Embodied-State Summary

experimental heuristic · fragmenting

Heuristic Participant Profile

experimental heuristic · needs-reinforcement-label

Awareness	71.1 %
Attention	70.8 %
Control	58.8 %
Continuity	70.2 %

Prototype fatigue indicator	**44.2%**
Latency sensitivity	**high**
Recovery strategy	**haptic reinforcement**

Reality Synchronization Engine

experimental composite · drifting

Robot	59.4 %
VR world	77.9 %
Body state	66.1 %
Unified	73.9 %

Narrative Summary Module

experimental · rule-based · fragmenting

Heuristic state label: fragmenting. The nominal descriptive rule condition flags latency for review.

- presence continuity is 70.2%
- dominant explanatory signal: latency
- rule status: nominal

Risk Rule Monitor

1 active

Medium embodiment degradation: agency or ownership is below the prototype rule threshold

Hypothesis Prompt Generator

rule-based prompt

Hypothesis: Visual-haptic mismatch may be associated with reduced ownership.

Test this using counterbalanced visual-haptic mismatch conditions with synchronized ownership ratings.

Session Pattern Summary

experimental heuristic · 76% rule-activation score

latency spike

ownership drift

adaptive compression

haptic reinforcement

Explanation Trace

experimental heuristic · 2 active factors

latency: latency exceeds the prototype rule threshold	**32%**
packet_loss: packet loss exceeds the prototype rule threshold	**8%**

Core Metrics

2026-05-22T10:00:00

Task time	**130 s**
Error rate	**11 %**
Heart rate	**106 bpm**
HRV	**42 ms**
Latency	**82 ms**
FPS	**61**
Packet loss	**1.3 %**

Relationships

2 sessions

Monitor	Candidate pattern: higher latency co-occurs with lower agency in current fixtures
Descriptive Pattern	Candidate descriptive pattern: higher embodiment and task performance co-vary in current fixtures
Descriptive Pattern	Candidate descriptive pattern: higher workload co-occurs with more errors in current fixtures

Research Analytics

descriptive correlations with sample size

latency vs agency	**insufficient n (2)**
embodiment vs task_efficiency	**insufficient n (2)**
workload vs error_rate	**insufficient n (2)**
heart_rate vs error_rate	**insufficient n (2)**

Experimental Research Architecture

29 modules

1. human operator
2. VR/AR/robot interface
3. multimodal sensor fusion layer
4. real-time cognitive state engine
5. embodiment intelligence model
6. adaptive telepresence system
7. research analytics and AI copilot

Session and Fixture Runs

descriptive comparison

Session ID	Task	Latency	Errors	Prototype summary
P-001	Remote object assembly	118	6	69.3
P-002	Remote navigation	82	11	65.7